



A New Measure of the Financial Cycle: Application to the Czech Republic

Miroslav Plašil, Jakub Seidler & Petr Hlaváč

To cite this article: Miroslav Plašil, Jakub Seidler & Petr Hlaváč (2016) A New Measure of the Financial Cycle: Application to the Czech Republic, Eastern European Economics, 54:4, 296-318, DOI: [10.1080/00128775.2016.1192950](https://doi.org/10.1080/00128775.2016.1192950)

To link to this article: <https://doi.org/10.1080/00128775.2016.1192950>



Published online: 19 Jul 2016.



Submit your article to this journal [↗](#)



Article views: 156



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 1 View citing articles [↗](#)

A New Measure of the Financial Cycle: Application to the Czech Republic

Miroslav Plašil

Financial Stability Department, Czech National Bank, Prague, Czech Republic

Jakub Seidler

ING Bank NV, Prague, Czech Republic

The Institute for Economic Studies, Charles University, Prague, Czech Republic

Petr Hlaváč

Financial Stability Department, Czech National Bank, Prague, Czech Republic

The recent financial crisis has demonstrated the importance of the linkages between the financial sector and the real economy. This article proposes a suitable and easy-to-apply method for assessing the position of the economy in the financial cycle in order to identify emerging imbalances in a timely manner. The method utilizes a composite indicator, constructed by the authors, that combines variables representing risk perceptions in the financial sector and their reinforcing interactions over the financial cycle. The indicator is calibrated to capture the future credit losses of the Czech banking sector. This method can be used by policymakers for a wide range of policy decisions, including the setting of a countercyclical capital buffer.

Keywords: Bayesian model averaging, countercyclical capital buffer, credit risk, financial cycle

JEL Classification: C11, E32, E37, E58

Crises and deadlocks, when they occur, have at least this advantage, that they force us to think.

—Jawaharlal Nehru

Miroslav Plašil is senior economist at the Financial Stability Department, Czech National Bank, Prague, Czech Republic.

Jakub Seidler is chief economist at the ING Bank NV, Prague, Czech Republic and was at the Financial Stability Department, Czech National Bank, Prague, Czech Republic at the time this paper was written.

Petr Hlaváč is not currently affiliated with an institution, but was senior economist at the Financial Stability Department, Czech National Bank, Prague, Czech Republic at the time this paper was written.

The views expressed in this article are those of the authors and not necessarily those of the Czech National Bank.

Correspondence should be addressed to Miroslav Plašil, Czech National Bank, CNB, Na Příkopě 28, 115 03, Prague 1, Czech Republic. E-mail: miroslav.plasil@cnb.cz

Color versions of one or more of the figures in the article can be found online at www.tandfonline.com/meece.

Recent developments in the global economy have contributed to the reassessment and elaboration of some well-established economic notions. Arguably, one of the most conspicuous changes in economic thinking is related to the greater importance of the linkages between the real and financial sectors. Due to the growing importance of financial intermediation, these linkages might lead to significant spillovers from one part of the economy to another and trigger substantial feedback loops. The milestones of the Great Recession may well illustrate their negative potential: a crisis initially linked with the housing market and its financing turned into an economic crisis and then a debt-and-banking crisis, which in turn significantly limited the scope for economic recovery. Facing this negative experience, economists have been forced to pay closer attention to the role of financial factors in business fluctuations and their impact on the overall soundness of the economic system. As a consequence, macroprudential instruments and financial stability issues have become an object of central interest to policymakers (e.g., Borio 2010, 2012; Borio and Drehmann 2009b).

From the macroprudential perspective, the reinforcing interactions between the real economy and the financial sector must be taken on board when analyzing emerging risks and vulnerabilities. However, even if business and financial fluctuations are related in some way, one should still be mindful of the differences in their nature and timing. The foundations of financial risks and imbalances are laid in good economic times, when expectations are running high. An expansionary phase of the financial cycle, associated with high (or even excessive) credit growth, is often followed by a deterioration in borrowers' ability to repay, growth in nonperforming loans, and large losses in the banking sector, which together can limit banks' ability to lend to the sound part of the real economy (Borio, Furfine, and Lowe 2001; Frait and Komárková 2012).

Against this backdrop, correctly determining the current phase of the financial cycle is vital for successfully identifying emerging risks, taking timely preventive action, and implementing stabilization policies. In practice, however, this objective may pose some practical challenges, as the financial cycle is mainly a theoretical quantity with no generally accepted empirical counterpart (e.g., Babecký et al. 2013; Borio and Drehmann 2009a). Despite the greater focus on the financial cycle in the literature in the years after the financial crisis, no definite answers have been provided, and a policy-relevant indicator for the financial cycle is still an open problem in the current policy debate. Although some may favor a parsimonious description of the financial cycle defined only in terms of one or very few variables (e.g., Borio 2012), relevant features of the cycle are echoed in far more indicators, which may provide some additional refinements. When assessing the current situation, policymakers can look at the whole set of individual indicators to get an idea of the cycle in action, but for a more formal quantitative assessment, as well as for communication purposes, it might still be useful to transform the existing information into a single representative measure.

Using data on the Czech economy, this article proposes a new indicator useful for monitoring developments over the financial cycle. Right at the beginning, it must be stressed that the ultimate *raison d'être* is to assist policymakers in activating and setting the countercyclical capital buffer (CCB). This goal largely predetermines the modeling choices in the article. In particular, the composite indicator should mainly capture those cyclical risks which can be effectively handled by the CCB, and closer attention is paid to the identification of those phases of the cycle where the need for macroprudential action is the most urgent (i.e., mainly the build-up phase of the cycle).

Conceptually, the discussion in this article follows previous articles that extracted cyclical information from a variety of financial variables (see Hatzius et al. 2010; Ng 2011). However, the empirical approach of this article is quite different from all previous financial cycle literature, and the composite indicator presented below is new in the financial cycle context. The novelty

lies in the aggregation method used, as well as in a more sensible approach to variable selection. The results suggest that the proposed measure of the financial cycle fully corresponds to the retrospective expert view on cyclical developments and mimics the changes in risk perceptions in the Czech economy very well. This is relevant in a situation where the apparent macro-prudential policy “star on the stage”—the HP-filtered credit/GDP gap—delivers highly erratic signals for the Czech economy.¹ Moreover, the proposed measure shows promising predictive power with respect to future credit losses, which makes it a useful guide for CCB-related decisions. The article is structured as follows. The next section briefly reviews related literature. The third section presents the motivation, description, and assessment of the composite indicator of systemic stress (CISS), which aggregates selected subindicators in a CISS-like fashion (see Holló, Kremer, and Lo Duca 2012). Thereafter, the predictive performance of the proposed financial cycle indicator is briefly sketched out with respect to GDP growth and credit risk materialization, followed by concluding remarks.

LITERATURE REVIEW

Historically, the financial cycle is considerably less well documented than business fluctuations. For most of the postwar period, economic research was practically silent in this regard. Mainstream economics saw the financial system as an amplifier of business fluctuations, just slightly complicating matters at best (e.g., Bernanke, Gertler, and Gilchrist 1998), while models that ignored financial variables completely were considered a relatively good approximation of reality. One of the few exceptions to the mainstream style was the work by Minsky (1982).

The main impetus for incorporating the financial sector into models (such as DSGE) was the occurrence of the financial crisis in the late 2000s. It turned out that the financial system can be at the center of a worldwide downturn and one of the direct causes of subsequent events. The financial crisis made it clear that the research that had been done so far on the identification of the financial cycle and its effects on the real economy was hardly sufficient.

The financial cycle literature, however, has some notable predecessors. In particular, it is important to highlight the work done on early warning indicators. These indicators were constructed to anticipate various kinds of crises, such as currency, debt, or banking crises. Since the peaks of financial cycles often coincide with the severest crises, early-warning indicators can naturally serve the same purpose as the identification of financial cycles. Krugman (1979) developed early-warning indicators to predict currency and balance-of-payments crises. Kaminsky and Reinhart (1999) and Reinhart and Rogoff (2011) proposed indicators to foresee debt and banking crises. The global financial crisis reawakened interest in this type of analysis, leading to the application of new empirical methods and larger and more detailed panels (e.g., Babecký et al. 2013; Laeven and Valencia 2010). The above-mentioned studies generally find that indicators showing changes in risk aversion (such as the rate of growth in credit to the private sector, debt and debt-servicing ability, property price growth, the tightness of the credit conditions, and the current account deficit or government debt level) are suitable indicators of future crises.²

The stream of research focusing on the procyclicality of the financial system, which can lead to unexpected boom and bust cycles and amplify economic fluctuations, can also be seen as a close relative of the financial cycle literature. This includes, for example, the articles by Adrian

and Shin (2010), Borio, Furfine, and Lowe (2001), Kashyap and Stein (2004), and White (2006). These studies, however, stayed more or less on the theoretical level and did not provide an unequivocal definition of the financial cycle.

The above theoretical studies are echoed in the empirical work of authors trying to capture the financial cycle with a particular variable or set of variables. This branch of research has grown to a considerable size. The most parsimonious way of defining the financial cycle empirically, but perhaps incompletely, is to use credit aggregates. For example, Dell’Ariccia et al. (2012) classify an episode as a credit boom simply if (a) the annual growth rate of the credit-to-GDP ratio exceeds 10% and the deviation from trend is greater than 1.5 times its standard deviation, or (b) the annual growth rate of the credit-to-GDP ratio exceeds 20%. The authors use bank credit in the numerator of the credit-to-GDP ratio and argue that it should be enough to capture the cycle (with one important exception, namely, the United States). Under this approach, one variable (although calculated from two time series) is sufficient to identify potentially dangerous upswings in the financial cycle; hence financial cycle is simply a different name for credit cycle. Similar studies in this respect include Aikman, Haldane, and Nelson (2010), Mendoza and Terrones (2008), and Schularick and Taylor (2012).

To obtain a more realistic picture, single-variable measures of the cycle were later enriched with other supplementary information, in particular that contained in property prices (Borio 2012; Drehmann, Borio, and Tsatsaronis 2012). Borio (2012) argues that credit developments together with property prices represent analytically the smallest set needed to replicate the mutually reinforcing interaction between financing constraints (as represented by credit) and perceptions of values and risks (as represented by property prices). These two quantities tend to move together, mainly at low frequencies, and seem to be fairly good leading indicators of banking crises, especially if used together. This reflects the fact that the credit gap is a rough measure of leverage in the economy, which indicates its loss-absorption capacity, and the property price gap indicates the probability and size of the potential price reversal that is symptomatic of financial crises. Drehmann, Borio, and Tsatsaronis (2012) also claim that recessions following upon the peaks of financial cycles tend to be more severe. In this sense, the financial cycle can be thought of as a leading indicator of the business cycle.

The financial cycle, as outlined above, still exploits only a very limited range of information. On the other side of the imaginary axis, there are methods based on factor models, which frequently utilize more than a dozen time series to identify the cycle. The variables are most often recruited from the area of market prices, lending conditions, or quantities related to interest rates (English, Tsatsaronis, and Zoli 2005; Hatzius et al. 2010; Ng 2011). Observed correlations between variables can be interpreted as the existence of a common latent factor that forces the variables to co-move. This factor, in turn, can be seen as a good proxy for the financial cycle in action.

A CISS-LIKE FINANCIAL CYCLE INDICATOR

Motivation and Design

The primary motivation for developing the composite indicator (hereinafter the FCI) was the need for a very simple indicator, one that would be very simple to construct, easy to interpret, and well understood by the widest possible audience. Experience showed that “rocket-science”

indicators usually fail to hit the intended target, since it is difficult to convey successfully their message to the relevant economic agents. In a similar vein, simpler indicators have a better chance of being employed as a communication vehicle and accepted by the general public. The growing popularity of the CISS indicator in the field of systemic risk measurement is grounds for some optimism that its underlying methodology goes in the right direction and is fairly easy to understand. The authors try to demonstrate that this does not come at the cost of losing good descriptive properties with respect to the financial cycle (its build-up phase in particular).

When proposing an indicator of a cycle, it is first necessary to explain what exactly it is to capture. As demonstrated in the preceding section, there is hardly a consensus definition of the financial cycle. The present article sticks to the definition adopted by the Czech National Bank in its Financial Stability Reports (for a more detailed account, see Frait and Komárková 2012, pp. 13–14). This definition is very close in spirit to the one in Borio (2012), for example. In what follows, the financial cycle is understood to mean recurrent swings in market participants' attitudes to financial risks, where the swings reflect changes in risk perceptions and value and the reinforcing interactions between them. Falling risk aversion is usually reflected in rapid credit growth, rising asset prices, easy access to external financing, and increased investment activity. Conversely, the downward phase of the cycle is accompanied by financing constraints and possible deleveraging in all sectors. Financial developments close to the peak and the trough of the financial cycle can considerably amplify business cycle fluctuations.

Using the above definition, the process of constructing the FCI can be split into two steps. The first step involves selecting relevant variables that capture changes in perceptions of financial risk across various segments of the economy. In the second step, the variables are combined into a single indicator using a simple aggregation algorithm. The main focus is placed on timely identification of the build-up phase of systemic risk and its subsequent shift to materialization, because this is the period when macroprudential intervention is most likely. The idea behind the FCI is to first analyze developments in particular segments of the economy and then make a synthesis in the form of a composite indicator. Note that the synthesis should be made only after a thorough analysis of individual information has been performed. This process thus allows for identification of common movements in the selected indicators while retaining knowledge of idiosyncratic developments in some segments, which may also call for some corrective policy action.³ The idiosyncratic nature of signals, however, may limit the scope of application of the countercyclical capital buffer, and other macroprudential measures may be preferred in this case.

Variable Selection

A researcher who has opted for one of the existing financial cycle concepts should not step out of the outlined territory. In this light, the main criterion for considering a variable a good candidate is that its information content should be in line with the definition of the financial cycle provided above. The authors' reading of the definition suggests that attention should be directed mainly at variables which monitor the *build-up* rather than the *materialization* of risks. Risk materialization indicators usually lag behind the cycle, as outlined above, or are even completely inverted in phase, as they often attain their most optimistic levels in the risk accumulation phase.⁴

From this perspective, the variables characterizing the financial cycle can be viewed as a set of forward-looking indicators of potential problems in the economy. Indeed, as already suggested, studies on early-warning systems generally find that variables connected with changes in risk perceptions—such as the rate of growth of credit to the private sector, the debt and debt-servicing ability of the private sector, property prices, the tightness of credit conditions, and the current account deficit or government debt level—are suitable indicators of future financial crises (e.g., Babecký et al. 2013; Leaven and Valencia 2010; Reinhart and Rogoff 2011).

The variables used to identify the Czech Republic's position in the financial cycle largely correspond to those listed in the above studies, taking into consideration data availability and quality.⁵ Every effort was made to cover the widest possible area of the economy that might be affected by changes in risk appetite, such as the credit demand and supply sides, property prices, debt sustainability, and general financial market sentiment. Wherever it makes sense, the input variables are compiled separately for the nonfinancial corporations sector and the household sector to make it easy to distinguish between sectoral tendencies and tendencies at the whole-economy (whole-private-sector) level.

A list of the input variables together with the adjustments made to them can be found in Table 1. The ranking of the variables in the table reflects the authors' expert assessment of their relevance to the identification of the individual phases of the financial cycle (arguably subjective but quite in line with the literature), but it also takes into account the quality of the data.⁶ A brief rationale for including each variable in the composite indicator is given in the following paragraphs.

Evolution of (New) Loans to Households and Nonfinancial Corporations

Many studies have shown that excessive credit growth is one of the best explanatory variables for future problems in the financial sector (see Babecký et al. 2013; Borio and Drehmann 2009a). This fact is linked with the procyclicality of the financial sector, as economic agents usually display a decreasing ability to recognize risks at times of economic growth and optimistic expectations.

TABLE 1
Definition of Input Indicators

<i>Indicator</i>	<i>Original unit and adjustments made</i>
1 New bank loans to households	CZK bn, annual moving sum of monthly new loans
2 New bank loans to nonfinancial corporations	CZK bn, annual moving sum of monthly new loans
3 Property prices (inflation)	y-o-y change in price index
4 Household debt/gross disposable income	bank loans/moving annual total, y-o-y change, %
5 Nonfinancial corporations' debt/gross operating surplus	bank loans/moving annual total, y-o-y change, %
6 Spread between rate on new loans to households and 3 M PRIBOR	% p.a., computed from quarterly average rates
7 Spread between rate on new loans to NFCs and 3 M PRIBOR	% p.a., computed from quarterly average rates
8 PX stock index	three-month average
9 Adjusted current account deficit/GDP	% p.a., adjusted for reinvestment and transfers

Source: CNB and CZSO, authors' calculations.

Faced with the prospect of rising future incomes, both households and firms are more willing to borrow. Analogously, lenders may be more willing (or may be exposed to strong incentives) to lend to riskier clients. The amount of new bank loans in a given period is used as an indicator of credit growth. Unlike the year-on-year change in the stock of loans, this indicator is not affected by the exclusion of bad loans from banks' balance sheets or by regular repayments of existing loans. The sizable structural break in the stock of loans within the data sample was also the main reason why indicators such as the credit-to-GDP ratio are not included in the FCI.⁷

Property Prices (Changes in the Property Price Index)

Many studies consider property market imbalances—associated with sharp growth in residential and commercial property prices—to be a factor that accompanies, or significantly accelerates, the onset of most financial crises (e.g., Allen, and Rogoff 2011; Drehmann, Borio, and Tsatsaronis 2012; Giese et al. 2014). Cheap financing in an optimistic phase of the financial cycle can push demand and prices above a sustainable level. The growth in prices can stimulate further credit expansion as a result of rising collateral value and the income effect on consumers (Bernanke, and Gertler 1995). The return to equilibrium is usually accompanied by negative effects on banks' balance sheets and by investment pessimism. As in Ng (2011), this article uses the year-on-year change in the property price index to capture imbalances in the property market (the index tracks property transaction prices as monitored by the CZSO on the basis of tax returns).

Debt Sustainability (Ratio of Household Debt to Gross Disposable Income, Ratio of Nonfinancial Corporation Debt to Gross Operating Surplus)

Rapid growth in the ratio of household debt to gross disposable income can signal that economic agents are overestimating their future ability to repay their debts. Higher growth in debt than in disposable income means that households may spend an increasingly large proportion of their income in the future on repaying their loans. If their income situation turns out worse than they expected, they will often become insolvent. The relationship between households' debt-to-disposable-income ratio and credit risk is described, for example, by Rinaldi and Arellano (2006).

A similar line of reasoning applies to the ratio of debt to gross operating surplus of nonfinancial corporations. In this case, moreover, the aspect of debt repayment sustainability is magnified by the fact that firms' total profits, which can be affected by variable or one-off items, do not figure in the denominator. In a converging economy the relative debt level of the private sector is constantly rising, so in this case falling risk aversion is measured using year-on-year changes, i.e., using the rate of growth of debt relative to income. Owing to the short time series available, total debt is proxied by bank loans only. However, as bank loans are the main source of external financing of the real sector, the informative value of these indicators should still be high.

Lending Conditions

Lending conditions characterize financial risk perceptions on the credit supply side and feature among the suitable indicators of future crises (Giese et al. 2014). In the growth phase

of the cycle, banks may encourage less creditworthy and more risky clients to borrow by offering low interest rates, but they have a tendency to underestimate the level of risk involved. In the risk materialization phase, by contrast, banks may tighten their lending conditions too much, leading to perceptible constraints on the financing of the sound part of the real sector (a credit crunch). As the bank lending survey in the Czech Republic has too short a history, the lending conditions are approximated using the difference between the interest rate on new loans to households/nonfinancial corporations and the three-month interbank rate (PRIBOR). Plašil, Radkovský, and Řežábek (2013) demonstrate that this simple approximation reproduces the results of the euro area survey quite reliably.

Stock Index (PX)

Some studies (e.g., Borio 2012) indicate that equity price volatility is not necessarily linked directly with the financial cycle, as it is determined more by the business cycle. However, the stock index may complete the overall picture of the nature of market participants' expectations and reveal over-optimism about future asset prices.

Adjusted Current Account Deficit-to-GDP Ratio

A current account deficit can be interpreted as meaning that more is invested in the economy than the private sector and the government save together. This may indicate the formation of external imbalances, overheating of the economy, and growth in future problems related to repaying loans financed by capital inflows from abroad (Giese et al. 2014). The current account also contains the income balance, which, in turn, has a reinvested earnings item. Countries which in past years attracted high FDI inflows (such as the Czech Republic) may face rising current account deficits due to growth in their income deficits. If, however, such deficits are driven by reinvested earnings, the growth is rather optical and does not mean a worsening external imbalance, because the reinvested earnings return to the host economy in the form of FDI. In other words, this is a relatively safe (though not entirely risk-free) source of capital. The current account also contains the balance of current transfers, which is determined primarily by the government sector income item (which often reflects random administrative factors). For this reason, the ratio of the current account to GDP was adjusted for the balance of reinvested earnings and the balance of transfers so as not to be distorted by these factors.

FCI—the Composite Indicator

Before aggregation, the input variables are first transformed into the unit interval (0, 1) using the kernel estimate of the cumulative distribution function (Gaussian kernel). The lowest value of the transformed variable corresponds to the trough of the cycle, and the highest value to the peak (see Figure 1).⁸ This step represents a form of standardization of the input variables, making their values mutually comparable. Mapping to the unit interval also facilitates subsequent interpretation of the FCI, as it provides a clearer idea about what is a low value and what is a high one. Note, however, that the transformation is based on historical distributions and as such is sample dependent.⁹ As new data arrive, the historical quantiles may change considerably, which, in turn, implies undesirable *ex post* revisions of past FCI values in real-time applications.

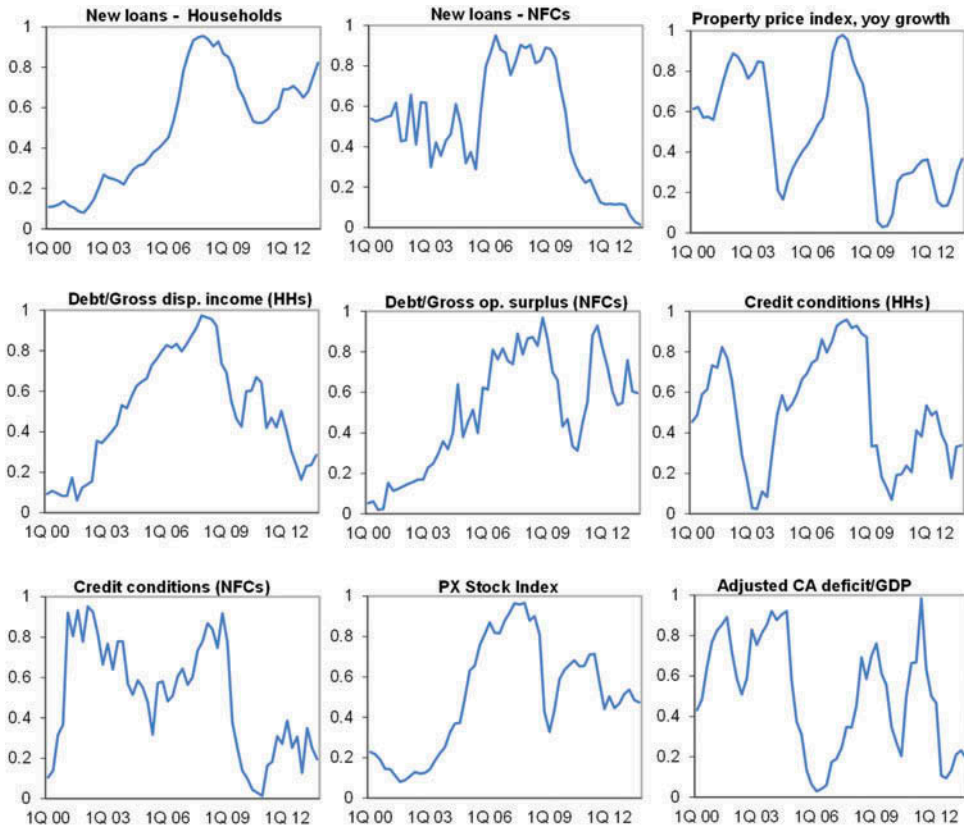


FIGURE 1 Input Variables after Their Mapping on Interval (0,1).

While this may raise some concerns in general, it is not considered a serious issue in the present setting, mainly because the chief interest here is in monitoring *future* developments. Given that the sample covers the last credit expansion and the subsequent return to the trough, it is argued that all future values are well anchored by the sample.¹⁰ However, even if revisions to the past values of the FCI were to occur in future, the comparison of the actual value with those around the last peak of the cycle would still bear valuable information.

Once the variables have been transformed into the unit interval, the aggregation method can be expressed using the formula in Equation (1) (see Holló, Kremer, and Lo Duca 2012)

$$FCI_t = (w^\circ s_t)' C_t (w^\circ s_t), \quad (1)$$

where a vector of weights, $w = (w_1, w_2, \dots, w_9)$, indicates the relative importance of the individual variables (subindicators), $s_t = (s_{1,t}, s_{2,t}, \dots, s_{9,t})$ is the vector of subindicators at time t , and $(w^\circ s_t)$ represents the element-by-element multiplication of these vectors (also known as the Hadamard product). Matrix C_t contains the values of the pairwise correlation coefficients

$\rho_{t,ij}$ determining how strong the relationship between subindicator i and j is at time t . Using aggregation (1), the result is a composite indicator defined on the interval (0, 1). The higher the indicator is, the higher is the degree of financial risk tolerance generally observed among market participants in the economy.

One of the main features of the chosen aggregation method is that it takes account of the time-varying cross-correlation structure of the data. The FCI generally takes higher values when variables are rising across all monitored segments. The stronger are the correlations between all the transformed variables (subindicators), the stronger is the signal sent out by the FCI about overall changes in sentiment over the cycle. This property is useful from the macroprudential perspective, particularly for setting the countercyclical capital buffer. As discussed in previous sections, the latter should be imposed in the event of general growth in cycle-related risks. If the growth is due to only some of the monitored segments (e.g., only growth in mortgage loans to households), it may be more appropriate to use a different prudential tool to eliminate the nascent risks.

In addition to the cross-correlation structure, which characterizes the interactions between individual segments and thus offers a cross-sectional view of the risks, the resultant aggregation captures the time dimension of risk.¹¹ The latter is given by the magnitude of the subindicators themselves. Their differing importance can be reflected in the resultant aggregation using a system of weights w .

It is useful to demonstrate the properties of the FCI using a simplified example. If the composite indicator were based on the aggregation of just three subindicators, its resultant value could be written as found in Equation (2):

$$\begin{aligned}
 FCI_t = & (w_1 s_{t,1} + w_2 \rho_{t,12} s_{t,2} + w_3 \rho_{t,13} s_{t,3}) w_1 s_{t,1} + \\
 & + (w_1 \rho_{t,12} s_{t,1} + w_2 s_{t,2} + w_3 \rho_{t,23} s_{t,3}) w_2 s_{t,2} + \\
 & + (w_1 \rho_{t,13} s_{t,1} + w_2 \rho_{t,23} s_{t,2} + w_3 s_{t,3}) w_3 s_{t,3}
 \end{aligned} \tag{2}$$

$\underbrace{\hspace{10em}}$
 Weight characterizing cross-
 sectional dimension of
 financial risk

$\underbrace{\hspace{10em}}$
 Weight characterizing time
 dimension of financial risk

It is clear from Equation (2) that the total weight of a subindicator in the FCI is given not only by the weights w themselves, but also by the value of the expression in parentheses, which, in turn, depends on the magnitude of the correlations between the given subindicator and other variables. If, for example, subindicator s_3 is not correlated with indicators s_1 and s_2 , its contribution to the FCI will be lower. Thus, a lower correlation represents a negative contribution to the final value of the FCI.

A special case is the situation where the correlation between all the subindicators is equal to one (perfect correlation), so that the FCI attains its upper bound with respect to the values of the subindicators. Comparing the current value of the composite indicator with its hypothetical maximum helps determine the extent to which the imperfect correlation structure reduces the FCI value; it enables calculation of the “signal loss” implied by imperfect synchronization of individual signals over the cycle. The overall value of the FCI can therefore be broken down into the positive contributions given by the subindicator values and the negative contribution (loss)

that depends on the cross-correlation structure of the data. The lower are the observed correlations between the subindicators, the larger is the negative contribution (in absolute terms).

Equation (2) shows that a set of variables that are strongly positively correlated with each other will have a larger positive effect on the final value of the FCI. In other words, variables exhibiting strong co-movement will (other things being constant) contribute most to the value of the FCI. On the contrary, its value can be substantially reduced when developments across segments are mixed and there is no clear signal about the current direction of risk perceptions. This behavior can be loosely interpreted as meaning that the FCI captures the effect of the latent factor causing the variables to co-move upwards.

Speaking of latent factors, the FCI may offer several advantages over the factor models commonly used to gather information on the co-movement of variables:

- With the short time series typical of most transforming economies, including the Czech Republic, it is difficult to verify (or ensure) the validity of the statistical assumptions needed to estimate factor models. The construction of the FCI may be less problematic in this regard.
- The output in the form of the FCI is more intuitive in nature and more intuitive to interpret, so it is more suitable for communication purposes. The proposed technique makes it easy to break down the indicator into the contributions of individual components and the effect of the correlations between the variables.
- A factor model (or the rather frequently used principal components) seeks to maximally reproduce the variability of the original data with a smaller number of artificial variables. However, this objective does not take into account whether the estimated factors display good predictive properties for some selected variable. By contrast, using an estimated correlation structure, the FCI makes it possible to set the time weights, w , on the variables optimally (in some sense), for example with the objective of best estimating future loan losses. In the case of the FCI, variables that have high loadings on the factor (e.g., the first principal component) can have a minimal weight if they do not help explain the materialization of credit risk.
- Basic factor models usually assume a constant cross-correlation structure over time, and hence constant relationships between the variables. In the case of the FCI, by contrast, the identification of changes in the cross-correlation structure is an important output, as it helps in identifying the individual phases of the financial cycle and reveals the formation of nonlinearities.

FCI—Calibration and Results for the Czech Economy

The Czech data relevant for the FCI do not have too long a history and suffer from structural breaks and lower reporting standards at the beginning of the sample. Constructing the FCI from the time series for the period 2000Q1–2013Q3 offers a reasonable compromise between data length and data quality (see [Figure 1](#)). To suppress the effect of the convergence of the Czech economy, variables that display constant upward growth trends due to a low initial level are expressed as year-on-year changes (see [Table 1](#) for the adjustments made).

Synchronization of variables during the cycle can be demonstrated using historical quantiles. The left panel of [Figure 2](#) shows the number of variables exceeding the 80th quantile in the given period,

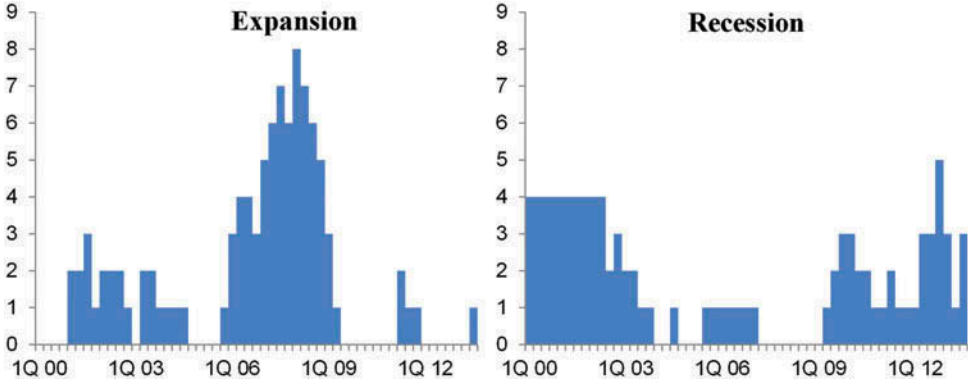


FIGURE 2 Simple Descriptive Measures of Co-movement.

Note: Expansion is measured as the number of indicators exceeding the 80th quantile. By contrast, recession is gauged by the number of indicators below the 15th quantile.

and the right panel depicts the number of variables below the 15th quantile. These quantities might serve as a simple measure of co-movement in the expansionary and recessionary phases, respectively. The dynamic nature of the relation between the variables can be readily observed, pointing to a need to model this feature explicitly. Within the FCI methodology, this phenomenon is formally captured by time-varying correlations.

The time-varying correlation coefficients were estimated recursively using the exponentially weighted moving average (EWMA) method with smoothing factor $\lambda = 0.94$ (RiskMetrics 1996). If the covariance σ_{ij} and variance σ_i^2 (or σ_j^2) at time $t - 1$ are known, the correlation coefficient $\rho_{t,ij}$ can be approximated using the following formulas:

$$\sigma_{t,ij} = \lambda \sigma_{t-1,ij} + (1 - \lambda) \tilde{s}_{t,i} \tilde{s}_{t,j}$$

$$\sigma_{t,i}^2 = \lambda \sigma_{t-1,i}^2 + (1 - \lambda) \tilde{s}_{t,i} \tilde{s}_{t,i}$$

$$\rho_{t,ij} = \sigma_{t,ij} / (\sigma_{t,i} \sigma_{t,j})$$

where, in line with Holló, Kremer, and Lo Duca (2012), $\tilde{s}_{t,i} = (s_{t,i} - 0.5)$ denotes the values of the individual subindicators after subtracting their “theoretical” median. The initial values of the correlation coefficients at time $t = 1$ were also estimated using the EWMA method applied to the time series in reverse order from the most recent observation to the oldest.¹²

The easiest way to calibrate vector w would be to assign equal weights to each subindicator. However, with regard to the intended use of the FCI for setting the countercyclical buffer, it seemed better to opt for a more targeted approach that incorporates prior expert knowledge and calibrates the weights of the FCI with respect to its predictive performance in terms of future credit losses. Expert knowledge is expressed in terms of *a priori* constraints on the vector of weights, which can be captured by the inequality in Equation (3):

$$w_1 \geq w_2 \geq w_3 \geq \dots \geq w_9, \quad (3)$$

where the subscripts correspond to the indicator ranking in Table 1. These constraints were imposed in order to reflect current literature and avoid unintuitive results. Note that the authors' *a priori* (expert) assessment of the relevance of each variable does not rule out the possibility that all the variables have the same weight.

The final calibration of the vector was determined by means of simulation techniques. A total of 30,000 different weight distributions satisfying Equation (3) were simulated; the vector chosen was the one that, after substitution into Equation (1), gave the best predictions of loan-loss impairments in the Czech banking sector six quarters ahead (in terms of RMSE).¹³ The chosen number of quarters reflects the fact that when a nonzero countercyclical capital buffer is announced, banks need at least one year to implement it. To this period, it is also necessary to add the data publication lag and the time needed to make the decision to set the capital buffer.

The estimated weights¹⁴ $w = (0.35 \ 0.27 \ 0.09 \ 0.08 \ 0.07 \ 0.05 \ 0.05 \ 0.02 \ 0.02)$ indicate that credit dynamics provide the main signal for forecasting the materialization of financial risks, as loans to households and nonfinancial corporations together have a weight of over 60% in the composite FCI. Using the estimated weights w and the correlations C_t , the FCI values can be obtained according to Equation (1). Figure 3 shows the evolution of the FCI (the black line) along with its decomposition into individual contributions (the bar chart).

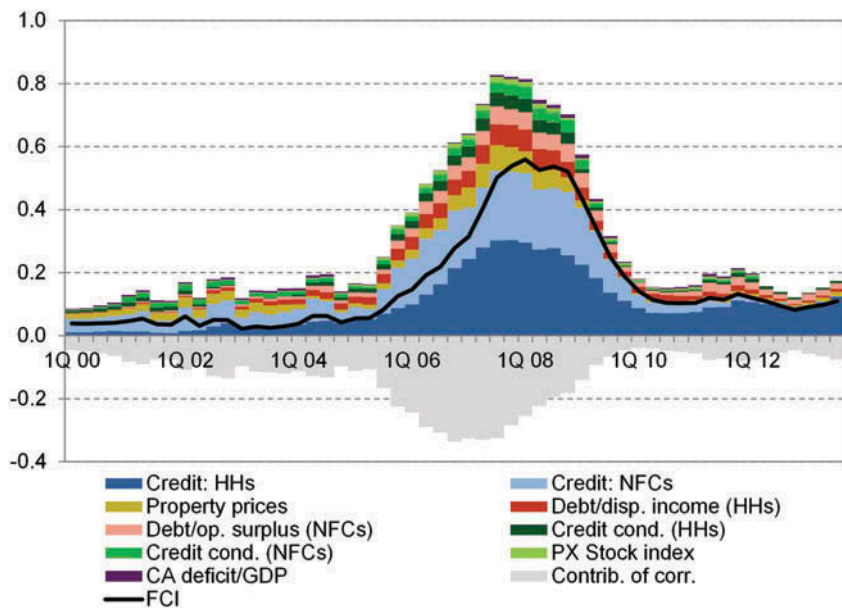


FIGURE 3 The FCI and its Decomposition.
Note: Minimum FCI = 0, maximum = 1. The negative contribution of the cross-correlation structure to the FCI (the loss due to imperfect correlation of the subindicators) is due to the difference between the current FCI value and the potential upper bound. Highly negative contributions indicate a generally weak correlation between the subindicators, whereas near-zero contributions indicate growing interconnectedness in individual areas of financial risk.

The results show that the FCI was very low until roughly the end of 2005. This reflected high financial risk aversion linked with the late-1990s banking crisis and the subsequent consolidation of the banking sector, which took until the start of the new millennium to complete. The period of 2005–2008 can be described as an expansionary phase of the financial cycle, with an economic recovery accompanied by gradually rising optimism and risk tolerance. Among other things, the expansion was fostered by the growing popularity of mortgage loans along with quite a strong construction boom and growth in property prices.¹⁵ In this period, bank clients showed a greater willingness to borrow despite the risks associated with future debt service. As time went on, this willingness was also fostered by the banks themselves through ever weaker lending conditions. Late 2008/early 2009 can be identified as the peak of the cycle. This was followed by a rapid switch to a downward phase of the cycle as a result of (the effects of) the financial crisis impacting on the Czech economy. The latest figures indicate that the Czech economy has been at the bottom of the financial cycle for some time now and is not showing any signs of accumulation of cyclical risks.¹⁶

The evolution of the FCI from the perspective of the cross-correlation structure and its contributions suggests that in the initial expansion phase (i.e., roughly between 2005 and 2007) the individual subindicators displayed quite mixed trends and the overall correlation between them was relatively low. This hindered growth in the composite indicator and manifested itself in a large difference between the upper bound and the actual value of the FCI (see [Figure 3](#)). By contrast, the peak phase of the cycle (2008/2009) was accompanied, in addition to growth in the contributions of the individual subindicators, by overall growth in the pairwise correlations, pushing the FCI value upwards and providing a stronger signal of risk accumulation. The correlations were still rising during the acute phase of the recession, when all the subindicators were falling together.

Overall, the path of the FCI indicator closely matches the retrospective views of experts on cyclical developments and seems to capture global changes in risk perceptions fairly well. To demonstrate its potential in the domain of macroprudential policy, it is worth comparing it with the benchmark macroprudential indicator for setting the countercyclical capital buffer rate, namely, the HP-filtered credit/GDP gap with lambda set to 400,000 (see [Figure 4](#)). One can see that the latter is highly misaligned with the generally accepted course of the cycle even if one accounts for the “starting point problem” as proposed in Drehmann and Tsatsaronis (2014).¹⁷ Given the length of the available sample, it can hardly be applied in Czech macroprudential practice. On the contrary, the FCI provides a very promising output. This issue is studied more formally in the next section, where its predictive properties are analyzed.

WHAT THE PROPOSED CYCLICAL MEASURE CAN SAY ABOUT FUTURE DEVELOPMENTS

Although the main purpose of the designed measure is different from forecasting, its predictive content with respect to some relevant quantities may be a positive side-effect. As stated in Ng (2011), if measures turn out to perform well for a policy-relevant purpose other than that for which they were constructed, then so much the better. This section provides some elementary insights on this subject. In particular, it analyzes the predictive performance of the proposed

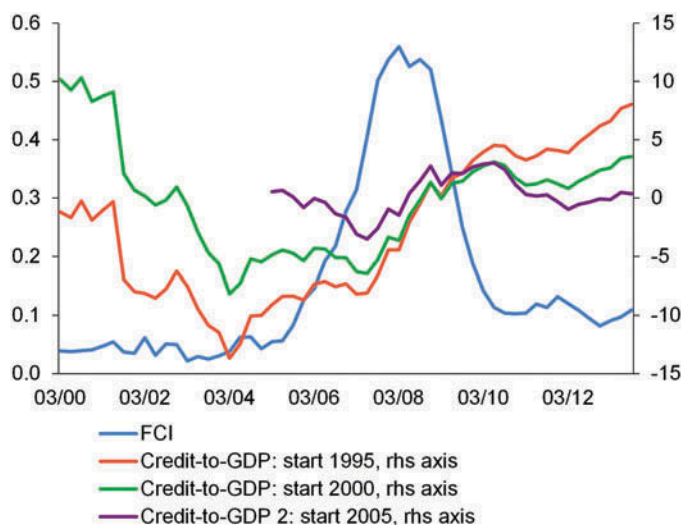


FIGURE 4 Comparison of the FCI and Credit-to-GDP Gaps.

Note: The HP filter with smoothing constant 400,000 was used to obtain the credit-to-GDP gap. Three different starting dates were considered.

financial cycle measure with respect to growth in GDP and nonperforming loans (as representatives of developments in the real economy and credit risk, respectively).

The issue of how financial developments amplify fluctuations in the real economy has frequently been investigated (e.g., Claessens, Kose, and Terrones 2012; Havránek, Horváth, and Matějů 2010; Ng 2011), so it is natural to explore this option here. In addition, the predictive content is analyzed with respect to growth in NPLs, since a fair estimate of its future path may provide a relevant quantitative link between the risk build-up phase and the size of the subsequent materialization. This might be particularly useful for decisions on the appropriate policy reaction (e.g., the setting of the appropriate countercyclical buffer rate).¹⁸

It should be noted that the aim is *not* to obtain the best possible predictions of GDP and NPL growth on the market. Strictly speaking, what is proposed is not even a forecasting exercise in the real sense, because the estimated values of the FCI are based on the whole data set and would have taken different values had they been calculated in real time.¹⁹ The following analysis is thus meant as a quick demonstration of whether the inclusion of the FCI can help explain future developments in the variables under study. To this end, attention is limited to a suite of very simple single-equation prediction models, which can be expressed by Equation (4):

$$q_{t+h} = \beta c_t + \delta X_t + \varepsilon_t, \quad (4)$$

where q_{t+h} is the variable of interest (either GDP growth or NPL) predicted at horizon h , c_t is a measure of the financial cycle, and X_t is a set of additional regressors. In this exercise, the horizon h is set to six quarters, as this is the time necessary for effective use of the countercyclical buffer and also roughly corresponds to the monetary policy horizon. For the sake of

simplicity, only the lags of the variable q are considered as additional regressors X_t (up to the fourth lag). Given this setting, the *de facto* interest here is in finding out whether the inclusion of the financial cycle measure can bring some additional gains over the information content in the past values of the forecasted variable. This question is answered by using two closely related methods, namely, Bayesian model averaging and dynamic model averaging.

The in-sample fit is assessed by means of Bayesian model averaging (BMA; readers unfamiliar with the methodology are referred to Hoeting et al. 1999 and Koop 2003). As a starting point, it is argued that there is considerable prior uncertainty about which model is the “correct” one for predicting q (i.e., GDP or NPL growth). In particular, we do not know whether it should contain the financial cycle measure ($\beta \neq 0$) or not ($\beta = 0$), and, similarly, it is not clear whether the financial cycle measure should enter the model separately ($\delta = 0$) or in combination with at least some lagged values of the variable q (some elements of δ are nonzero). It is assumed that a comprehensive set of all admissible models is formed as any combination of the measure of the financial cycle and four lags of q , and that all models are a priori equally likely. The key output of BMA is a posterior model probability for each model considered, which can be interpreted as the probability that the given model is the “correct” one given the observed data. Aggregation of posterior probabilities over all models containing the financial cycle measure then leads to its posterior inclusion probability—the probability that the measure is included in the model. The BMA results for both GDP and NPL growth are given in Table 2 and Figure 5.

One can clearly see that the posterior inclusion probability (PIP) of the FCI measure virtually equals one in both cases, delivering strong evidence of its good in-sample performance against purely autoregressive models. It is also apparent that the coefficient β has the same sign across various model specifications, which is economically appealing (see Figure 5). By contrast, the coefficients δ frequently change their signs across lags and models, making their interpretation

TABLE 2
BMA Estimates for Model (4)

	PIP	Posterior BMA Mean	Posterior BMA SD	Pos. sign certainty
<i>Forecasted variable: HDP</i>				
(Pseudo $R^2 = 0.31$)				
IFC _{<i>t-6</i>}	1,00	-10,77	2,66	0,00
GDP _{<i>t-6</i>}	0,27	-0,01	0,12	0,30
GDP _{<i>t-7</i>}	0,30	0,03	0,16	0,78
GDP _{<i>t-8</i>}	0,32	0,00	0,18	0,52
GDP _{<i>t-9</i>}	0,53	0,15	0,21	1,00
<i>Forecasted variable: NPL</i>				
(Pseudo $R^2 = 0.92$)				
IFC _{<i>t-6</i>}	1,00	82,36	4,53	1,00
NPL _{<i>t-6</i>}	0,09	0,01	0,04	1,00
NPL _{<i>t-7</i>}	0,09	0,00	0,04	0,06
NPL _{<i>t-8</i>}	0,08	0,00	0,04	0,87
NPL _{<i>t-9</i>}	0,08	0,00	0,02	0,02

Note: PIP stands for posterior inclusion probability; positive sign certainty (last column) indicates the probability that the coefficient on the given variable takes a positive value.

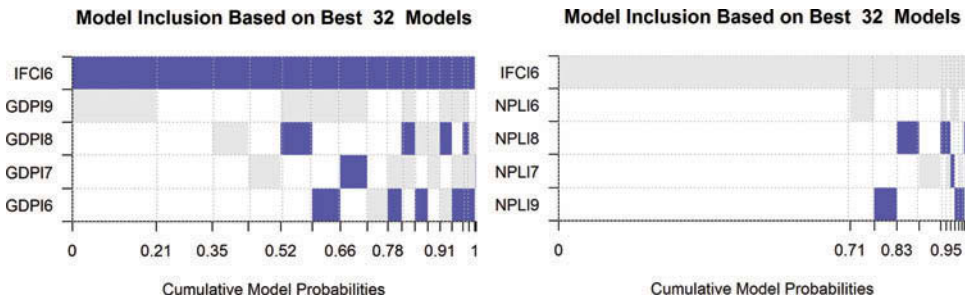


FIGURE 5 Cumulative Posterior Model Probabilities for NPL and GDP Growth.

Note: The x-axis shows the cumulative model probabilities of the best 32 models. The models are ordered according to their posterior probabilities, with the best model being on the left (individual models are separated by dashed lines). Lighter color indicates a positive coefficient and darker color a negative coefficient, while white color denotes the absence of the variable from the given model.

far less intuitive. The overall fit of the model (measured by pseudo R^2 ²⁰) is quite impressive for NPL growth, suggesting that the proposed financial cycle measure may be very useful in explaining credit-risk materialization. In the case of GDP growth, the overall predictive performance seems to be quite unsatisfactory even after the inclusion of the financial cycle measure. Besides the fact that it can be difficult to forecast GDP for longer horizons (especially when recession threats are *ante portas*), this may also point to nonlinearities in the relation between real and financial developments. Recent experience suggests that they are usually little intertwined in normal times, but their interplay may be substantial around the peak of the financial cycle.

The hypothesis of potential nonlinearities is explored a little further by using the method of dynamic model averaging (DMA; see Raftery, Kárný, and Ettler 2010 for technical details). It can be seen as an extension of traditional BMA into a time-varying framework, where the parameters β and δ as well as the posterior model probabilities are allowed to change in each time period. As different models can now be preferred in different periods, DMA represents a natural candidate for studying the nature of the nonlinearities outlined above. Another feature of DMA is that it uses filtered estimates of β_t and δ_t , which brings it closer to the real-time forecasting exercise. The filtered values of the parameters at time t are based solely on the information available up to time t . Note that this is very different from the situation analyzed before (the BMA estimates), where the fixed coefficients were estimated using the whole sample. In the case of BMA the “in-sample” fit was analyzed, whereas DMA focuses on the one-step-forward “out-of-sample” performance.

Figure 6 depicts the changes in the posterior model probabilities across time. To reduce visual clutter, it only shows the aggregate posterior probability for models containing the financial cycle measure vis-à-vis the probability for models without it (effectively, this is the posterior inclusion probability of the cyclical measure vis-à-vis its complement to unity). As is immediately apparent, in the case of NPL growth (left panel), models containing the financial cycle

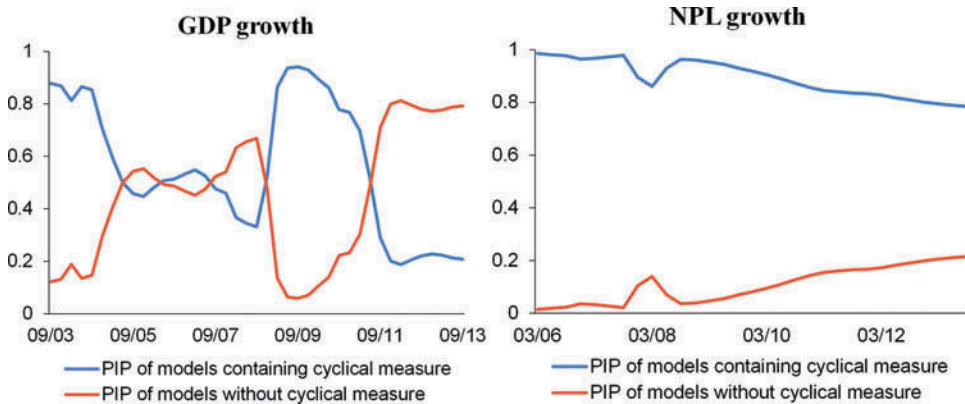


FIGURE 6 Time-varying Posterior Inclusion probabilities for the FCI.
Note: The time axis is constructed from the perspective of the forecasted variable. For example: the high posterior probability for GDP growth in 2009Q3 means that models containing a measure of the financial cycle can generate a better prediction for this period (using information lagged by six quarters, i.e., information up to 2008Q1).

measure convincingly dominate their competitors over the entire sample. This suggests that the measure might be useful for predicting future changes in credit risk in expansionary as well as recessionary phases of the cycle. By contrast, in the case of GDP growth (right panel), the posterior probabilities exhibit a markedly different pattern. As can be seen, the probability of inclusion of the financial cycle measure reaches its peak in the late expansionary phase (in the lead-up to recession) and the subsequent recessionary phase. This means that past financial developments contribute considerably to explaining business cycle fluctuations at this time. By contrast, during other periods the models with the cyclical measure performed roughly as well (or as poorly) as the models containing only lagged values, hence they do not show any comparative gains. In the recent past, which can be characterized as a “crawling along the trough” of the financial cycle, they even seem to perform worse than autoregressive models. These results are interpreted as evidence of a nonlinear relation between financial and business fluctuations, a relation that tends to be strong around the tipping point of the financial cycle and is quite loose otherwise.

Overall, the proposed financial cycle measure withstands the forecasting challenge quite well. If used cautiously, it may provide helpful guidance for macroprudential decisions and also, to lesser extent, for monetary policy decisions. In the latter case this is chiefly true for periods when the financial cycle starts to amplify business fluctuations.

CONCLUSION

This article focuses on modeling of the financial cycle in the Czech economy. While the recent financial crisis has highlighted the importance of relations between business and financial fluctuations, finding a policy-relevant indicator for the financial cycle is still an open problem in the

current policy debate. Using data on the Czech economy, this article proposes an easy-to-apply method for measuring the financial cycle with respect to the setting of the countercyclical buffer.

The proposed method uses a set of variables measuring swings in risk perceptions and aggregates them into a single composite indicator. One of the main features of the aggregation method is that it takes into account both the co-movement of variables and idiosyncratic developments. Since time-varying cross-correlation structure of the data is employed, the resulting indicator generally takes higher values when the variables reach elevated levels in all monitored segments and the risk build-up signal is stronger.

In selecting the employed variables, the authors covered the widest possible area of the economy that might be affected by changes in risk perceptions (i.e., the credit demand and supply sides, property prices, debt sustainability, general financial market sentiment, and external imbalances). Alongside the time-varying cross-correlation structure, the resulting financial cycle indicator (FCI) employs a system of fixed weights assigned to each variable. These weights were calibrated so as to provide optimal predictive performance for loan-loss impairment six quarters ahead. Timely identification of risk materialization is necessary for making timely decisions about the CCB setting.

The results confirm that the period of 2005–2008 can be described as an expansionary phase of the financial cycle, with an economic recovery accompanied by gradually rising risk tolerance. In this period, bank clients showed a greater willingness to borrow despite the risks associated with future debt service. As time went on, this willingness was also fostered by the banks themselves through ever weaker lending conditions. Late 2008/early 2009 can be identified as the peak of the cycle. This was followed by a rapid switch to a downward phase of the cycle as a result of the financial crisis impacting on the Czech economy. Unlike for the credit-to-GDP gap, these developments closely correspond to economic intuition and are in line with current expert judgment. In this light, the measures may better serve macroprudential purposes than the traditionally used credit-to-GDP gap and may provide policymakers with a useful framework for assessing the financial cycle.

An initial analysis of the predictive content of the cyclical measures suggests that they may contribute to a more precise assessment of future credit-risk materialization in both the expansionary and the recessionary phase, and to some extent, they may also help predict developments in the real economy, notably around a tipping point of the financial cycle. This seems to confirm earlier literature on the observed nonlinearities in the relation between macroeconomic and financial developments.

ACKNOWLEDGMENTS

The authors would like to thank Mathias Drehmann, Jakub Matějů, Štefan Rychtárik, and Roman Horváth for their helpful comments. All remaining errors are our own.

FUNDING

This work was supported by Czech National Bank Research Project No. C4/13 and by Czech Science Foundation Grant No. P402/12/G097.

Notes

1. Credit-to-GDP measures can deliver highly misleading signals in some countries, as shown in Geršl and Seidler (2011). This is particularly true when the credit aggregate is polluted by structural breaks, as is the case in the Czech economy.
2. Forward-looking indicators also differ across studies, depending on the set of countries examined. For emerging economies, foreign exchange reserves and the equilibrium real exchange rate are often appropriate indicators (e.g., Frankel and Rose 1996).
3. In contrast to common factor models, idiosyncratic developments should not automatically be seen as noise, because they may still contain valuable information from the macroprudential perspective.
4. We would argue that bundling together risk accumulation and risk materialization indicators lacks theoretical interpretation and cannot result in any meaningful measure.
5. The choice of variables would be rather different for other countries where the data quality is higher and longer time series for relevant indicators are available. Some very useful variables (e.g., the debt service ratio) could not be used in the present case, since the time series necessary for calculating them start only in 2004.
6. This ranking is used later in simulating the weights of the input variables in the resulting composite indicator.
7. On the other hand, figures for new credit can be tainted by the phenomenon of loan refinancing—the situation where one bank takes over a client's debt from another bank to enlarge its credit portfolio (refinancing becomes part of the “new credit aggregate” despite the fact that no additional credit was extended to the sector in economic terms). Since the available data do not allow the series to be adjusted for this phenomenon, one should take it into account when interpreting the final outcome.
8. The original CISS applies a rather simpler transformation using the empirical cumulative distribution function. Some of the variables (spreads, current account deficit/GDP) had to be multiplied by a coefficient of -1 before the transformation, so that low financial risk aversion corresponded to higher values for all the variables.
9. We are not the first to propose a measure of the financial cycle based on the comparison of indicators with their own history and subsequent aggregation; see Rychtárik (2014). However, the mapping and the aggregation algorithm (as well as our conceptual approach to the variable selection) are different in the present case.
10. Current experience seems to support this claim, as *ex post* revisions of the past values of the FCI were negligible when data for 2013Q4–2014Q4 were added to the sample (not reported).
11. Definitions of the time and cross-sectional dimension of risk can be found in Frait and Komárková (2012). In this article, however, the cross-sectional dimension of risk is defined rather differently. The original concept defines the cross-sectional dimension as the degree of financial interconnection between economic agents, which can generate financial risks, whereas here the cross-sectional dimension is taken to mean the degree of interconnection between the various aspects of financial risk, which can amplify the overall level of financial risk.
12. This is not necessarily a very precise approximation of the initial values, but we are mainly interested in the relative value of the correlations vis-à-vis their value in other periods, rather than their absolute value.
13. As an alternative, the weights were determined with regard to the predictive power of the FCI for the 12-month default rate in the nonfinancial corporations sector and for the first difference of the ratio of nonperforming loans to total loans in the private sector. The results were similar.
14. The weights are rounded and their order corresponds to that in Table 1. Rounded values were used to calculate the FCI.
15. Hlaváček and Komárek (2011) point to some overvaluation on the property market in 2007 and 2008. This reflected, among other things, a pre-announced increase in VAT on residential property construction.
16. In recent years, moreover, the FCI values have been further overestimated due to the phenomenon of mortgage refinancing, which is inflating the total amount of new loans to households. It is not yet possible to fully filter out this effect based on the available statistics.

17. As a rule of thumb, Drehmann and Tsatsaronis (2014) propose the use of credit gaps only when at least ten years of data are already available for the credit-to-GDP ratio. Another option is to drop the initial data points.
18. Note that the weights w in the FCI were already chosen optimally with respect to the predictions for loan losses six quarters ahead. As loan losses and NPL growth are clearly correlated, there is a sort of circularity in this exercise. However, *optimal* values (leading to the minimum RMSE) do not necessarily imply *good* predictions. Thus, they are still considered to show how *good* the obtained optimum is.
19. Splitting the sample into calibration and verification subsamples, however, would hardly have been possible in the given conditions due to the short time series available. To obtain reasonable estimates of the cycle, a span covering expansion and contraction phases is needed. Moreover, this article does not use the vintage data which were available to policymakers at time t .
20. Unity minus the posterior variance over the dependent variable.

REFERENCES

- Adrian, T., and H. S. Shin. 2010. "The Changing Nature of Financial Intermediation and the Financial Crisis of 2007–09." Staff Report No. 439, Federal Reserve Bank of New York, New York. https://www.newyorkfed.org/medialibrary/media/research/staff_reports/sr439.pdf
- Aikman, D., A. Haldane, and B. Nelson. 2010. "Curbing the Credit Cycle." Paper presented at the Columbia University Center on Capitalism and Society Annual Conference, New York, NY, November.
- Allen, F., and K. Rogoff. 2011. "Asset Prices, Financial Stability and Monetary Policy." Unpublished manuscript.
- Babecký, J., T. Havránek, J. Matějů, M. Rusnák, K. Šmídková, and B. Vašíček. 2013. "Leading Indicators of Crisis Incidence: Evidence from Developed Countries." *Journal of International Money and Finance* 35 (C):1–19. doi:10.1016/j.jimonfin.2013.01.001.
- Bernanke, B. S., and M. Gertler. 1995. "Inside the Black Box: The Credit Channel of Monetary Policy Transmission." *Journal of Economic Perspectives* 9 (4):27–48. doi:10.1257/jep.9.4.27.
- Bernanke, B.S., M. Gertler, and S. Gilchrist. 1998. "The Financial Accelerator in a Quantitative Business Cycle Framework." Working Paper No. 6455, National Bureau of Economic Research, Cambridge, MA. <http://www.nber.org/papers/w6455.pdf>
- Borio, C. 2010. "Implementing a Macroprudential Framework: Blending Boldness and Realism." Paper presented at the HKIMR-BIS Conference on Financial Stability: Towards a Macroprudential Approach, Hong Kong, July 5–6. <http://www.bis.org/repofficepubl/hkimr201007.12c.pdf>
- Borio, C. 2012. "The Financial Cycle and Macroeconomics: What Have We Learnt?" Working Paper No. 395, Bank for International Settlements, Basel, Switzerland. <http://www.bis.org/publ/work395.pdf>
- Borio, C., and M. Drehmann. 2009a. "Assessing the Risk of Banking Crises—Revisited." *BIS Quarterly Review* March 2009 :29–46. http://www.bis.org/publ/qtrpdf/r_qt0903e.pdf
- Borio, C., and M. Drehmann. 2009b. "Towards an Operational Framework for Financial Stability: Fuzzy Measurement and Its Consequences." Working Paper No. 284, Bank for International Settlements, Basel, Switzerland. <http://www.bis.org/publ/work284.pdf>
- Borio, C., C. Furfine, and P. Lowe. 2001. "Procyclicality of the Financial System and Financial Stability: Issues and Policy Options." In *Marrying the Macro- and Micro-Prudential Dimensions of Financial Stability*, published in *BIS Paper*, Bank for International Settlements, No. 1. 1–57. <http://www.bis.org/publ/bppdf/bispap01a.pdf>
- Claessens, S., M. A. Kose, and M. E. Terrones. 2012. "How Do Business and Financial Cycles Interact?" *Journal of International Economics* 87 (1):178–90. doi:10.1016/j.jinteco.2011.11.008.
- Dell’Ariccia, G., D. Igan, L. Laeven, H. Tong, B. Bakker, and J. Vandenbussche. 2012. "Policies for Macrofinancial Stability: How to Deal with Credit Booms." Staff Discussion Note 12/06, International Monetary Fund, Washington, DC. <https://www.imf.org/external/pubs/ft/sdn/2012/sdn1206.pdf>
- Drehmann, M., and K. Tsatsaronis. 2014. "The Credit-to-GDP Gap and Countercyclical Capital Buffers: Questions and Answers." *BIS Quarterly Review* March 2014:55–73. http://www.bis.org/publ/qtrpdf/r_qt1403g.pdf

- Drehmann, M., C. Borio, and K. Tsatsaronis. 2012. "Characterising the Financial Cycle: Don't Lose Sight of the Medium Term!" Working Paper 380, Bank for International Settlements, Basel, Switzerland. <http://www.bis.org/publ/work380.pdf>
- English, W., K. Tsatsaronis, and E. Zoli. 2005. "Assessing the Predictive Power of Measures of Financial Conditions for Macroeconomic Variables." *BIS Papers* No: 228–252. 22. <http://www.bis.org/publ/bppdf/bispap22n.pdf>
- Frait, J., and Z. Komárková. 2012. "Macroprudential Policy and Its Instruments in a Small EU Economy." Research and Policy Note 3/2012, Czech National Bank, Prague, Czech Republic. https://www.cnb.cz/miranda2/export/sites/www.cnb.cz/en/research/research_publications/irpn/download/rpn_3_2012.pdf
- Frankel, J. A., and A. K. Rose. 1996. "Currency Crashes in Emerging Markets: An Empirical Treatment." International Finance Discussion Papers No. 534, Board of Governors of the Federal Reserve System (U.S.), Washington, DC. <http://www.federalreserve.gov/pubs/ifdp/1996/534/ifdp534.pdf>
- Geršl, A., and J. Seidler. 2011. "Credit Growth and Capital Buffers: Empirical Evidence from Central and Eastern European Countries." Research and Policy Note 2/2011, Czech National Bank, Prague, Czech Republic. https://www.cnb.cz/miranda2/export/sites/www.cnb.cz/en/research/research_publications/irpn/download/rpn_2_2011.pdf
- Giese, J., H. Andersen, O. Bush, C. Castro, M. Farag, and S. Kapadia. 2014. "The Credit-to-GDP Gap and Complementary Indicators for Macroprudential Policy: Evidence from the UK." *International Journal of Finance & Economics* 19 (1):25–47. doi:10.1002/ijfe.v19.1.
- Hatzius, J., P. Hooper, F. Mishkin, K. Schoenholtz, and M. Watson. 2010. "Financial Conditions Indexes: A Fresh Look After the Financial Crisis." Working Paper No. 16150, National Bureau of Economic Research, Cambridge, MA. <http://www.nber.org/papers/w16150.pdf>
- Havránek, T., R. Horváth, and J. Matějů. 2010. "Do Financial Variables Help Predict Macroeconomic Environment? The Case of the Czech Republic." Working Paper 6/2010, Czech National Bank, Prague, Czech Republic. https://www.cnb.cz/miranda2/export/sites/www.cnb.cz/en/research/research_publications/cnb_wp/download/cnbwp_2010_06.pdf
- Hlaváček, M., and L. Komárek. 2011. "Regional Analysis of Housing Price Bubbles and Their Determinants in the Czech Republic." *Czech Journal of Economics and Finance* 61 (1):67–91. http://journal.fsv.cuni.cz/storage/1205_1205_hlavacek_upr.graf_10_resume.pdf
- Hoeting, J. A., D. Madigan, A. E. Raftery, and C. T. Volinsky. 1999. "Bayesian Model Averaging: A Tutorial." *Statistical Science* 14:382–401. https://projecteuclid.org/download/pdf_1/euclid.ss/1009212519
- Holló, D., M. Kremer, and M. Lo Duca. 2012. "CISS—A Composite Indicator of Systemic Stress in the Financial System." Working Paper No. 1426, European Central Bank, Frankfurt, Germany. <https://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp1426.pdf>
- Kaminsky, G. L., and C. M. Reinhart. 1999. "The Twin Crises: The Causes of Banking and Balance-of-Payments Problems." *American Economic Review* 89 (3):473–500. doi:10.1257/aer.89.3.473.
- Kashyap, A., and J. C. Stein. 2004. "Cyclical Implications of the Basel II Capital Standards." *Economic Perspectives* (28):18–31. https://www.chicagofed.org/digital_assets/publications/economic_perspectives/2004/ep_1qtr2004_part2_kashyap_stein.pdf
- Koop, G. 2003. *Bayesian Econometrics*. Chichester, UK: Wiley.
- Krugman, P. 1979. "A Model of Balance-of-Payments Crises." *Journal of Money, Credit and Banking* 11 (3):311–25. doi:10.2307/1991793.
- Leaven, I., and F. Valencia. 2010. "Resolution of Banking Crises: The Good, the Bad and the Ugly." Working Paper 10/146, International Monetary Fund, Washington, DC. <https://www.imf.org/external/pubs/ft/wp/2010/wp10146.pdf>
- Mendoza, E., and M. Terrones. 2008. "An Anatomy of Credit Booms: Evidence from Macro Aggregates and Micro Data." Working Paper No. 14049, National Bureau of Economic Research, Cambridge, MA <http://www.nber.org/papers/w14049.pdf>
- Minsky, H. 1982. *Can "It" Happen Again? Essays on Instability and Finance*. Armonk, NY: M.E. Sharpe.
- Ng, T. 2011. "The Predictive Content of Financial Cycle Measures for Output Fluctuations." *BIS Quarterly Review* June 2011:53–65. http://www.bis.org/publ/qtrpdf/r_qtl106g.pdf
- Plašil, M., Š. Radkovský, and P. Řežábek. 2013. "Modelling Bank Loans to Non-Financial Corporations." Thematic Article, Financial Stability Report 2012/2013, Czech National Bank, 128–36, Prague, Czech Republic. http://www.cnb.cz/en/financial_stability/fs_reports/fsr_2012-2013/fsr_2012-2013_article_v.pdf
- Raftery, A. E., M. Kárný, and P. Ettler. 2010. "Online Prediction Under Model Uncertainty via Dynamic Model Averaging: Application to a Cold Rolling Mill." *Technometrics* 52:52–66. doi:10.1198/TECH.2009.08104.
- Reinhart, C. M., and K. S. Rogoff. 2011. "From Financial Crash to Debt Crisis." *American Economic Review* 101 (5):1676–1706. doi:10.1257/aer.101.5.1676.

- Rinaldi, L., and A. Arellano. 2006. "Household Debt Stability: What Explains Household Non-Performing Loans? An Empirical Analysis." Working Paper 570, European Central Bank, Frankfurt, Germany. <http://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp570.pdf>
- RiskMetrics Group. 1996. *Technical Document*. New York, NY: MSCI. <https://www.msci.com/documents/10199/5915b101-4206-4ba0-ace2-3449d5c7e95a>
- Rychtárik, Š. 2014. "Analytical Background for the Counter-Cyclical Capital Buffer Decisions in Slovakia." *Biatec* 22 (4):10–15. http://www.nbs.sk/_img/Documents/_PUBLIK_NBS_FSR/Biatec/Rok2014/04-2014/03_biatec14-04_rychtarik.pdf
- Schularick, M., and A. M. Taylor. 2012. "Credit Booms Gone Bust: Monetary Policy, Leverage Cycles, and Financial Crises, 1870–2008." *American Economic Review* 102 (2):1029–61. doi:10.1257/aer.102.2.1029.
- White, W. R. 2006. "Procyclicality in the Financial System: Do We Need a New Macrofinancial Stabilization Framework?" Working Paper No. 193, Bank for International Settlements, Basel, Switzerland. <http://www.bis.org/publ/work193.pdf>